

SQL-Based Data Analysis: Query Outputs and Results

```
CREATE DATABASE superstore_db;
```

```
USE superstore_db;
```

```
-- 1. EXPLORE TABLE
```

```
-- View first 10 rows
```

```
SELECT * FROM superstore LIMIT 10;
```

Result Grid															Filter Rows:		Export:		Wrap Cell Contents:		Fetch rows:	
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID								
▶	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10								
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10								
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10								
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10									
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10									
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10									
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10									
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10									
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AT-10C									
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10									

-- Total number of records

```
SELECT COUNT(*) AS total rows FROM superstore;
```

A screenshot of the 'Result Grid' toolbar. It includes a 'Filter Rows' button with a funnel icon, an 'Export' button with a document icon, and a 'Wrap Cell Content' button with a text icon. Below the toolbar, a small table shows 'total_rows' with a value of 9694.

```
-- View table schema
```

DESCRIBE superstore;

Result Grid

Filter Rows:

Export:

Wrap Cell Content: [T](#)

	Field	Type	Null	Key	Default	Extra
▶	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

-- 2. WHERE FILTERS

-- Orders from West region

SELECT * FROM superstore WHERE region = 'West';

Result Grid														Filter Rows:	Export:	Wrap Cell Contents:	Fetch rows:
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product				
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-				
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-				
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-				
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-				
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-1				
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-				
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-				
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-				
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-1				
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-				
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-				
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-				
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-1				
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-				
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-1				
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-				
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-				
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC-				

-- Furniture category orders

SELECT * FROM superstore WHERE category = 'Furniture';

Result Grid		Filter Rows:		Export:		Wrap Cell Content:		Fetch rows:							
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases
30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings
37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furnishings
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Off...	United States	Houston	Texas	77041	Central	FUR-BO-10002545	Furniture	Bookcases
40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Off...	United States	Houston	Texas	77041	Central	FUR-CH-10004218	Furniture	Chairs
52	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer	United States	New Albany	Indiana	47150	Central	FUR-FU-10001706	Furniture	Furnishings
53	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer	United States	New Albany	Indiana	47150	Central	FUR-CH-10003061	Furniture	Chairs
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East	FUR-CH-10003968	Furniture	Chairs
66	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	FUR-FU-10000397	Furniture	Furnishings
67	US-2015-164175	4/30/2015	5/5/2015	Standard Class	PS-18970	Paul Stevenson	Home Off...	United States	Chicago	Illinois	60610	Central	FUR-CH-10001146	Furniture	Chairs
73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South	FUR-CH-10000513	Furniture	Chairs

-- Sales greater than 1000

SELECT * FROM superstore WHERE sales > 1000;

Result Grid		Filter Rows:		Export:		Wrap Cell Content:							
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	Sales	City	State	Postal Code	Region
1614	CA-2017-138380	12/21/2017	12/25/2017	Standard Class	YS-21880	Yana Sorensen	Corporate	United States	1000.02	Oakland	California	94601	West
6211	CA-2017-119809	8/18/2017	8/25/2017	Standard Class	YS-21880	Yana Sorensen	Corporate	United States	1000.02	Seattle	Washington	98103	West
5157	CA-2017-163006	6/30/2017	7/4/2017	Second Class	GH-14410	Gary Hansen	Home Off...	United States	1001.584	Chicago	Illinois	60653	Central
6908	CA-2016-140249	9/27/2016	10/3/2016	Standard Class	SW-20455	Shaun Wielen	Consumer	United States	1001.584	Seattle	Washington	98103	West
2482	CA-2017-147956	12/24/2017	12/31/2017	Standard Class	AH-10210	Alan Hwang	Consumer	United States	1003.62	Seattle	Washington	98105	West
409	CA-2017-117457	12/8/2017	12/12/2017	Standard Class	KH-16510	Keith Herrera	Consumer	United States	1004.024	San Francisco	California	94110	West
1506	CA-2015-153388	8/1/2015	8/7/2015	Standard Class	PC-19000	Pauline Chand	Home Off...	United States	1004.976	Los Angeles	California	90004	West
1631	CA-2016-106530	5/8/2016	5/8/2016	Same Day	CL-12565	Clay Ludtke	Consumer	United States	1006.056	Cleveland	Ohio	44105	East
949	US-2017-110576	11/28/2017	12/2/2017	Standard Class	RB-19795	Ross Baird	Home Off...	United States	1007.232	Philadelphia	Pennsylvania	19120	East
6579	CA-2014-103660	8/25/2014	8/30/2014	Standard Class	MW-18220	Mitch Webber	Consumer	United States	1007.944	Seattle	Washington	98103	West
437	CA-2016-147375	6/12/2016	6/14/2016	Second Class	PO-19180	Phillisse Overcash	Home Off...	United States	1007.979	Chicago	Illinois	60623	Central
7082	CA-2016-106950	9/2/2016	9/6/2016	Standard Class	JE-15715	Joe Elijah	Consumer	United States	1012.68	Charlotte	North Carolina	28205	South

-- Profit greater than 500

SELECT * FROM superstore WHERE profit > 500;

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	Profit	City	State	Postal Code
9166	CA-2016-100300	6/24/2016	6/26/2016	Second Class	MJ-17740	Max Jones	Consumer	United States	503.64	San Diego	California	92037
6296	CA-2014-141313	12/28/2014	1/1/2015	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	503.7822	Beverly	Massachus...	1915
1712	CA-2017-123491	10/30/2017	11/5/2017	Standard Class	JK-15205	Jamie Kunitz	Consumer	United States	510.636	San Francisco	California	94122
1251	CA-2017-115602	12/18/2017	12/24/2017	Standard Class	DJ-13630	Doug Jacobs	Consumer	United States	511.368	New York City	New York	10009
4882	CA-2017-143567	11/2/2017	11/5/2017	Second Class	TB-21175	Thomas Boland	Corporate	United States	517.4793	Henderson	Kentucky	42420
7488	US-2015-134558	12/19/2015	12/24/2015	Second Class	PM-19135	Peter McVee	Home Office	United States	517.4793	Dover	New Hamp...	3820
4429	CA-2017-119452	3/21/2017	3/27/2017	Standard Class	CL-12565	Clay Ludtke	Consumer	United States	523.7052	Tulsa	Oklahoma	74133
5517	US-2016-138408	11/18/2016	11/21/2016	Second Class	KC-16255	Karen Carlisle	Corporate	United States	527.984	Huntsville	Alabama	35810
9948	CA-2017-121559	6/1/2017	6/3/2017	Second Class	HW-14935	Helen Wasserman	Corporate	United States	539.2464	Indianapolis	Indiana	46203
3385	CA-2015-130659	12/4/2015	12/9/2015	Second Class	MS-17365	Maribeth Schnelling	Consumer	United States	539.925	New York City	New York	10035
9514	CA-2016-125220	10/14/2016	10/19/2016	Standard Class	BE-11410	Bobby Elias	Consumer	United States	544.4175	Appleton	Wisconsin	54915
7133	CA-2017-141439	11/26/2017	12/1/2017	Standard Class	TT-21460	Tonja Turnell	Home Office	United States	545.1884	Richmond	Indiana	47374

-- Orders placed in 2017

SELECT * FROM superstore WHERE `Order Date` LIKE '%2017';

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Pr
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OF
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FU
35	CA-2017-107727	10/19/2017	10/23/2017	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	77095	Central	OF
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United States	Naperville	Illinois	60540	Central	TB
44	CA-2017-139619	9/19/2017	9/23/2017	Standard Class	ES-14080	Erin Smith	Corporate	United States	Melbourne	Florida	32935	South	OF
72	CA-2017-114440	9/14/2017	9/17/2017	Second Class	TB-21520	Tracy Blumstein	Consumer	United States	Jackson	Michigan	49201	Central	OF
76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OF
78	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OF
85	US-2017-119662	11/13/2017	11/16/2017	First Class	CS-12400	Christopher Schild	Home Office	United States	Chicago	Illinois	60623	Central	OF

-- 3. AGGREGATION FUNCTIONS

-- Total Sales

SELECT SUM(sales) AS total_sales FROM superstore;

total_sales
2272449.8562999545

-- Total Quantity Sold

SELECT SUM(quantity) AS total_quantity FROM superstore;

total_quantity
36749

-- Average Sales

```
SELECT ROUND(AVG(sales),2) AS average_sales FROM superstore;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	average_sales			
▶	234.42			

-- Maximum Sale

```
SELECT MAX(sales) AS highest_sale FROM superstore;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	highest_sale			
▶	22638.48			

-- Minimum Sale

```
SELECT MIN(sales) AS lowest_sale FROM superstore;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	lowest_sale			
▶	0.444			

-- Total Profit

```
SELECT ROUND(SUM(profit),2) AS total_profit FROM superstore;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_profit			
▶	282857.75			

-- 4. GROUP BY ANALYSIS

-- Sales by Region

```
SELECT
```

```
    region,
```

```
    ROUND(SUM(sales),2) AS total_sales
```

```
FROM superstore
```

GROUP BY region

ORDER BY total_sales DESC;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
region	total_sales		
West	713471.34		
East	672194.05		
Central	497800.87		
South	388983.59		

-- Profit by Region

SELECT

region,

ROUND(SUM(profit),2) AS total_profit

FROM superstore

GROUP BY region

ORDER BY total_profit DESC;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
region	total_profit		
West	106021.15		
East	90672.01		
South	46035.69		
Central	40128.9		

-- Sales by Category

SELECT

category,

ROUND(SUM(sales),2) AS total_sales

FROM superstore

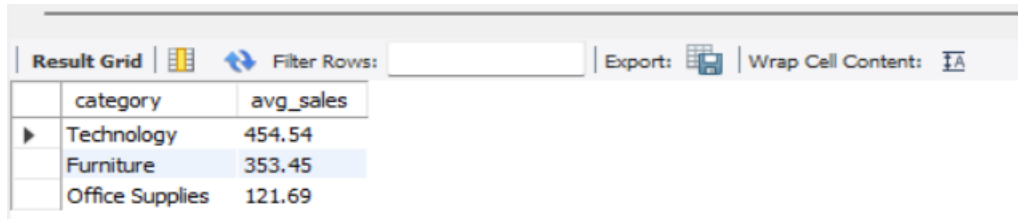
GROUP BY category

ORDER BY total_sales DESC;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category	total_sales		
Technology	835900.07		
Furniture	733046.86		
Office Supplies	703502.93		

-- Average Sales by Category

```
SELECT
    category,
    ROUND(AVG(sales),2) AS avg_sales
FROM superstore
GROUP BY category
ORDER BY avg_sales DESC;
```

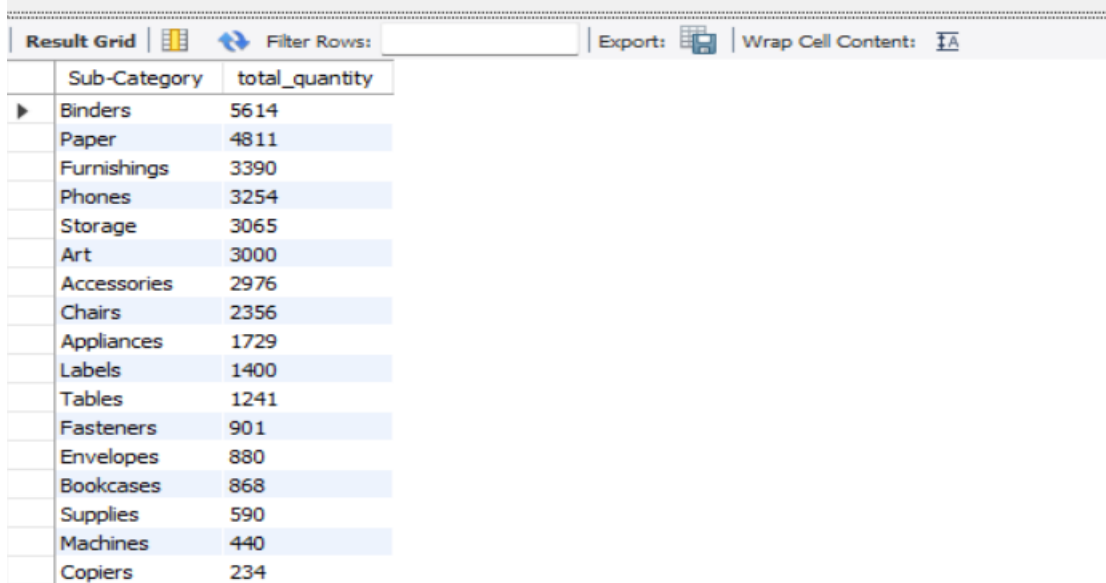


The screenshot shows a SQL query result grid with the following data:

	category	avg_sales
▶	Technology	454.54
	Furniture	353.45
	Office Supplies	121.69

-- Quantity Sold by Sub-Category

```
SELECT
    `Sub-Category`,
    SUM(Quantity) AS total_quantity
FROM superstore
GROUP BY `Sub-Category`
ORDER BY total_quantity DESC;
```



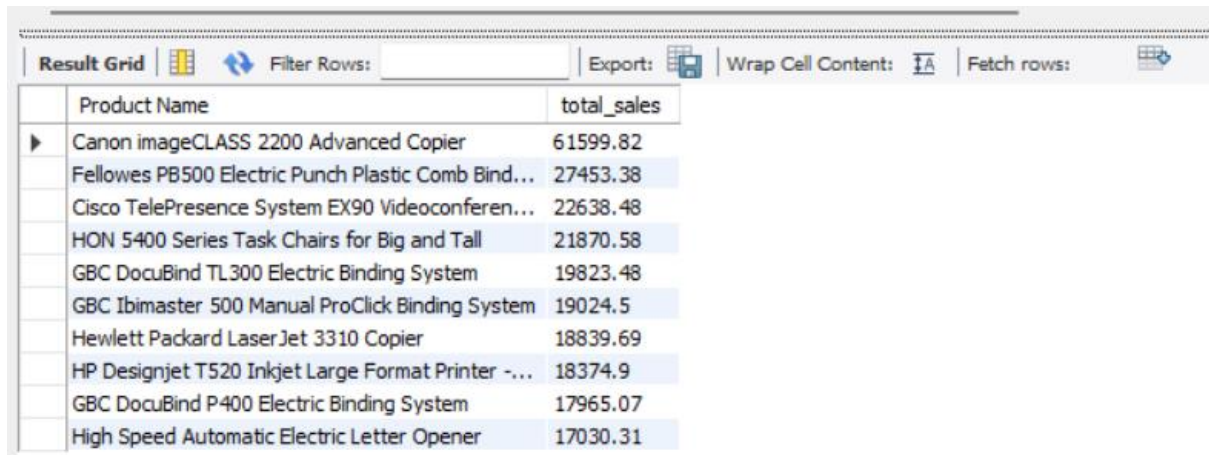
The screenshot shows a SQL query result grid with the following data:

	Sub-Category	total_quantity
▶	Binders	5614
	Paper	4811
	Furnishings	3390
	Phones	3254
	Storage	3065
	Art	3000
	Accessories	2976
	Chairs	2356
	Appliances	1729
	Labels	1400
	Tables	1241
	Fasteners	901
	Envelopes	880
	Bookcases	868
	Supplies	590
	Machines	440
	Copiers	234

-- 5. SORTING AND LIMIT

-- Top 10 Products by Sales

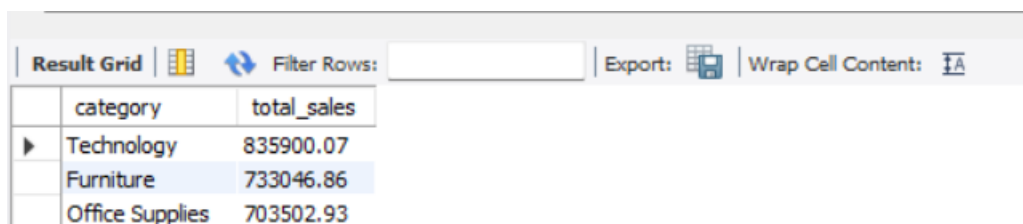
```
SELECT
    'Product Name',
    ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY 'Product Name'
ORDER BY total_sales DESC
LIMIT 10;
```



Product Name	total_sales
Canon imageCLASS 2200 Advanced Copier	61599.82
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.38
Cisco TelePresence System EX90 Videoconferen...	22638.48
HON 5400 Series Task Chairs for Big and Tall	21870.58
GBC DocuBind TL300 Electric Binding System	19823.48
GBC Ibimaster 500 Manual ProClick Binding System	19024.5
Hewlett Packard LaserJet 3310 Copier	18839.69
HP Designjet T520 Inkjet Large Format Printer -...	18374.9
GBC DocuBind P400 Electric Binding System	17965.07
High Speed Automatic Electric Letter Opener	17030.31

-- Top Categories by Sales

```
SELECT
    category,
    ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY category
ORDER BY total_sales DESC;
```



category	total_sales
Technology	835900.07
Furniture	733046.86
Office Supplies	703502.93

-- Top 10 Customers by Sales

```

SELECT
    `Customer Name`,
    ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY `Customer Name`
ORDER BY total_sales DESC
LIMIT 10;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	Customer Name	total_sales				
▶	Sean Miller	25043.05				
	Tamara Chand	19017.85				
	Raymond Buch	15117.34				
	Tom Ashbrook	14595.62				
	Adrian Barton	14355.61				
	Sanjit Chand	14142.33				
	Ken Lonsdale	14071.92				
	Hunter Lopez	12873.3				
	Sanjit Engle	12209.44				
	Christopher Conant	12129.07				

-- 6. BUSINESS USE CASES

-- Monthly Sales Trend

```

SELECT
    DATE_FORMAT(`Order Date`, '%Y-%m') AS month,
    ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY month
ORDER BY month;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	month	total_sales			
▶	NULL	2272449.86			

-- Monthly Profit Trend

```

SELECT
    DATE_FORMAT(`Order Date`, '%Y-%m') AS month,

```

```

ROUND(SUM(profit),2) AS total_profit
FROM superstore
GROUP BY month
ORDER BY month;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	month	total_profit			
▶	NULL	282857.75			

```

-- Top 5 Customers
SELECT
    `Customer Name`,
    ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY `Customer Name`
ORDER BY total_sales DESC
LIMIT 5;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	Customer Name	total_sales				
▶	Sean Miller	25043.05				
	Tamara Chand	19017.85				
	Raymond Buch	15117.34				
	Tom Ashbrook	14595.62				
	Adrian Barton	14355.61				

```

-- Most Profitable Products
SELECT
    `Product Name`,
    ROUND(SUM(profit),2) AS total_profit
FROM superstore
GROUP BY `Product Name`
ORDER BY total_profit DESC
LIMIT 10;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

IA

Fetch rows:

	Product Name	total_profit
▶	Canon imageCLASS 2200 Advanced Copier	25199.93
	Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.04
	Hewlett Packard LaserJet 3310 Copier	6983.88
	Canon PC1060 Personal Laser Copier	4570.93
	HP Designjet T520 Inkjet Large Format Printer -...	4094.98
	Ativa V4110MDD Micro-Cut Shredder	3772.95
	3D Systems Cube Printer, 2nd Generation, Mag...	3717.97
	Plantronics Savi W720 Multi-Device Wireless He...	3696.28
	Ibico EPK-21 Electric Binding System	3345.28
	Zebra ZM400 Thermal Label Printer	3343.54

-- Top 10 States by Sales

SELECT

state,

ROUND(SUM(sales),2) AS total_sales

FROM superstore

GROUP BY state

ORDER BY total_sales DESC

LIMIT 10;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	state	total_sales
▶	California	450567.59
	New York	309453.63
	Texas	169553.63
	Washington	136590.17
	Pennsylvania	114911.24
	Florida	88876.88
	Illinois	79009.29
	Ohio	76617.64
	Michigan	75991.58
	Virginia	70309.09

-- 7. DUPLICATE RECORD CHECK

SELECT

`Order Id`,

COUNT(*) AS duplicate_count

FROM superstore

```
GROUP BY `Order Id`
HAVING COUNT(*) > 1;
```

	Order Id	duplicate_count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2
	CA-2015-117415	4

-- 8. DATA QUALITY CHECKS

-- Validate total rows

```
SELECT COUNT(*) AS total_rows FROM superstore;
```

	total_rows
▶	9694

-- Check NULL values

```
SELECT
    COUNT(*) AS total_rows,
    COUNT(`Order Id`) AS order_id_count,
    COUNT(`Customer Name`) AS customer_name_count,
    COUNT(`Product Name`) AS product_name_count,
    COUNT(sales) AS sales_count
FROM superstore;
```

	total_rows	order_id_count	customer_name_count	product_name_count	sales_count
▶	9694	9694	9694	9694	9694

-- Missing customer names

SELECT * FROM superstore WHERE 'Customer Name' IS NULL;

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
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-- Missing product names

SELECT * FROM superstore WHERE 'Product Name' IS NULL;

Result Grid

Filter Rows:

Exports:

Wrap Cell Content:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
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-- Negative sales values

SELECT * FROM superstore WHERE sales < 0;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
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-- Negative profit values

SELECT * FROM superstore WHERE profit < 0;

Result Grid													Filter Rows:		Export:		Wrap Cell Content:		Fetch rows:			
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Profit	Country	City	State	Postal Code									
▶	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	-383.031	United States	Fort Lauderdale	Florida	33311									
	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	-123.858	United States	Fort Worth	Texas	76106									
	16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	-3.816	United States	Fort Worth	Texas	76106									
	24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	-1.0196	United States	Philadelphia	Pennsylvania	19140									
	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	-1665.0522	United States	Philadelphia	Pennsylvania	19140									
	29	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	-7.0532	United States	Philadelphia	Pennsylvania	19140									
	37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	-147.963	United States	Richardson	Texas	75080									
	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	-46.9764	United States	Houston	Texas	77041									
	40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	-16.147	United States	Houston	Texas	77041									